# D. K .T. E. Society's
# Textile and Engineering Institute, Ichalkaranji
## (An Autonomous Institute)

DKTE

AN AUTONOMOUS INSTITUTE

NAAC ACCREDITED WITH 'A +' GRADE

**Syllabus**

**for**

**Third Year of Computer Science and Engineering**

**(With effect from 2018-19)**

# Third Year B. Tech. Computer Science and Engineering Structure

D.K.T.E.Society's Textile and Engineering Institute,Ichalkaranji.
An Autonomous Institute
Teaching and Evaluation Scheme for
Third Year B.Tech. Computer Science & Engineering -Semester-V

| Sr. No. | Course Code | Name of the Course | Group | Teaching Scheme | | | | | | | | | | Credits |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Theory Hrs/ Week | Tutorial Hrs/ Week | Practical Hrs/ Week | Total | SE-I | SE-II | SEE | Practical CIE | Practical SEE | Total | |
| 1 | CSL301 | Operating systems-I | D | 3 | | - | 3 | 25 | 25 | 50 | - | - | 100 | 3 |
| 2 | CSL302 | Database Engineering | D | 3 | | - | 3 | 25 | 25 | 50 | - | - | 100 | 3 |
| 3 | CSL303 | Machine Learning | D | 3 | | - | 3 | 25 | 25 | 50 | - | - | 100 | 3 |
| 4 | CSL304 | Information Security | D | 3 | 1 | - | 4 | 25 | 25 | 50 | - | - | 100 | 4 |
| 5 | CSL305 | System Programming | D | 3 | - | - | 3 | 25 | 25 | 50 | - | - | 100 | 3 |
| 6 | CSP306 | Object Oriented Programming using Java | D | 3 | - | 4 | 7 | | - | - | 50 | 50 | 100 | 5 |
| 7 | CSP307 | Open Source Lab-I | D | - | - | 2 | 2 | | - | - | 50 | - | 50 | 1 |
| 8 | CSP308 | Database Engineering -Lab | D | | | 2 | 2 | | | | 50 | 50 | 100 | 1 |
| 9 | CSD309 | Seminar | F | - | 1 | | 1 | | - | - | 50 | - | 50 | 1 |
| | | **Total** | | **18** | **2** | **8** | **28** | **125** | **125** | **250** | **200** | **100** | **800** | **24** |
| 10 | CSI310 | Soft Skill(Audit Course)* | C | 2* | | | | | | | | | | Grade |
| | | **Total** | | **20** | **2** | **8** | **28** | **125** | **125** | **250** | **200** | **100** | **800** | **24** |

**Abbreviations:**

CIE: Continuous Internal Evaluation
SEE: Semester End Examination
ISE: In Semester Evaluation
SE-I: Semester Examination-I
SE-II: Semester Examination-II

**Group Details**

A: Basic Science
B: Engineering Science
C: Humanities,Social Science and Management
D: Professional Subjects-Core & Electives
E: Open Electives
F: Seminar/Training/Project

* Course for only pass grade

**D.K.T.E.Society's Textile and Engineering Institute,Ichalkaranji.**
**An Autonomous Institute**
**Teaching and Evaluation Scheme for**
**Third Year B.Tech. Computer Science & Engineering -Semester-VI**

| Sr. No. | Course Code | Name of the Course | Group | Teaching Scheme | | | | | | | | | | Credits |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Practical | | Total | |
| | | | | Theory Hrs/ Week | Tutorial Hrs/ Week | Practical Hrs/ Week | Total | SE-I | SE-II | SEE | CIE | SEE | | |
| 1 | CSL311 | Operating systems-II | D | 3 | - | - | 3 | 25 | 25 | 50 | - | - | 100 | 3 |
| 2 | CSL312 | Compiler Construction | D | 3 | 1 | - | 4 | 25 | 25 | 50 | - | - | 100 | 4 |
| 3 | CSL313 | Computer Algorithm | D | 3 | 1 | - | 4 | 25 | 25 | 50 | - | - | 100 | 4 |
| 4 | CSL314 | Data Science | D | 3 | - | - | 3 | 25 | 25 | 50 | - | - | 100 | 3 |
| 5 | OE | Open Elective | E | 3 | | | 3 | 25 | 25 | 50 | | - | 100 | 3 |
| 6 | CSP315 | Web Technologies | D | 3 | - | 4 | 7 | - | - | - | 50 | 50 | 100 | 5 |
| 8 | CSP316 | Open Source Lab II | D | - | - | 2 | 2 | - | - | - | 100 | - | 100 | 1 |
| 9 | CSD317 | Miniproject-II | F | - | - | 2 | 2 | - | - | - | 50 | 50 | 100 | 2 |
| 7 | CSP318 | Industrial Training / Internship | D | - | - | - | 0 | - | - | - | 100 | - | 100 | 2 |
| | | **Total** | | **18** | **2** | **8** | **28** | **125** | **125** | **250** | **300** | **100** | **900** | **27** |

| | | |
|---|---|---|
| | **Abbreviations:** | |
| | CIE: | Continuous Internal Evaluation |
| | SEE: | Semester End Examination |
| | ISE: | In Semester Evaluation |
| | SE-I: | Semester Examination-I |
| | SE-II: | Semester Examination-II |

**Open Elective**
ETLOE1 : Fundamentals of Embedded System
MELOE2 : Mechatronics
TFLOE1 : Merchandising

# Third Year B. Tech. Computer Science and Engineering – Semester – V

## Third Year B. Tech.
## CSL301: OPERATING SYSTEM-I

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| **Total** | **100** |

| Prerequisite |
|---|
| Computer Organization, Microprocessors |

| Course Objectives | |
|---|---|
| 1. | To learn basic concepts of operating system and its services. |
| 2. | To describe various features of processes including scheduling, creation and termination |
| 3. | To present critical section problem with its solutions and different methods for preventing or avoiding deadlocks in a computer system. |
| 4. | To discuss various memory management strategies and file system design. |
| 5. | To discuss goals and principles of protection and fundamentals of encryption, authentication and security defences. |

| Course Outcomes<br>At the end of the course students will be able to | |
|---|---|
| 1. | Describe the structure , functions and services of an Operating System |
| 2. | Apply process management algorithms to solve process related problems |
| 3. | Explain the memory management techniques in operating systems |
| 4. | Explain I/O management techniques in Operating Systems. |

| | Course Contents | |
|---|---|---|
| **Unit 1.** | **Introduction to OS and Services:-** Operating System fundamentals, computer system organization, computer system architecture, OS structure, OS operations, process management, memory management, storage management, computing environments.<br>OS services, user and OS interface, system calls, types of system calls, OS structure, system boot. | **7 Hrs.** |
| **Unit 2.** | **Process Management:-**Process concept, Process States, Process Control Block, inter-process communication, process scheduling:- basic concepts, Scheduling Criteria, Scheduling Algorithms, Multiple processor scheduling, Real time CPU scheduling. | **7 Hrs.** |
| **Unit 3.** | **Process Synchronization and Deadlocks:-**Background, The critical section problem, Peterson's solution, Mutex Locks, Semaphores, Classic problems of synchronization. System model, deadlock characterization, handling deadlocks, deadlock prevention, deadlock avoidance, deadlock detection, recovery from deadlock. | **6 Hrs.** |
| **Unit 4.** | **Memory Management:-**Background, swapping, contiguous memory allocation, segmentation, paging, structure of page table.<br>Virtual memory background, demand paging, copy-on-write, page replacement, allocation of frames, thrashing. | **7 Hrs.** |
| **Unit 5.** | **Storage Management:-**File concept, access methods, directory and disk structure, file system mounting, file sharing, protection. | **6 Hrs.** |
| **Unit 6.** | **Input / Output Systems:-**Overview, I/O hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance. | **6 Hrs** |

| **Text Books** | |
|---|---|
| 1. | Abraham Silberschatz, Peter B Galvin, Gerg Gagne "Operating System Concepts", 9th Edition. |
| 2. | William Stallings, Operating System: Internals and Design Principles, Prentice Hall, ISBN-10: 0-13-380591-3, ISBN-13: 978-0-13-380591-8, 8th Edition |

| Reference Books | |
|---|---|
| 1 | Operating system with case studies in Unix, Netware and Windows NT – Achyut S. Godbole (TMGH). |
| 2 | "Operating systems: concepts and design" - Milan Milenkovic (TMGH). |
| 3 | "Operating Systems: Internals and Design Principles" by William Stallings |

**Third Year B. Tech (Semester- V)**

## CSL302- Database Engineering

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | - |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Object Oriented Concepts, Fundamentals of Data Structures |

| Course Objectives | |
|---|---|
| 1 | To develop conceptual understanding of fundamentals of database management systems. |
| 2 | To get familiar with different data models. |
| 3 | To apply functional dependency and normalization. |
| 4 | To model the database by query writing. |
| 5 | To expose students to transaction processing and concurrency control mechanisms. |

| Course Outcomes<br>At the end of the course students will be able to - | |
|---|---|
| 1 | Explain the fundamentals of database management systems. |
| 2 | Design database using E-R Model. |
| 3 | Apply the principles and practices of good database design like functional dependency and normalization. |
| 4 | Devise queries using SQL. |
| 5 | Compare different indexing schemes. |
| 6 | List different concurrency control mechanisms. |

| Contents | | |
|---|---|---|
| **Unit 1** | **Introduction to DBMS :–**Introduction, Traditional file system v/s DBMS, views of data, instance and schema, Data Models – Relational and ER model, Keys, Database design process, Schema diagram.<br>Database system structure, Database users.<br>Relational algebra, Tuple relational calculus, Domain relational calculus. | **8 Hrs** |
| **Unit 2** | **Structured Query Language (SQL) :–** Introduction to SQL, data types.<br>DDL Statements – Create, Alter, Drop, Rename, Truncate. | **6 Hrs** |

| | | |
|---|---|---|
| | DML Statements- Select, Insert, Update, Delete.<br>DCL Statements – Commit, Rollback.<br>Joins, Group by-having clause, order by clause, aggregate functions, set operations, authorization statements – grant, revoke, Foreign key, String operations, Nested Queries. | |
| **Unit 3** | **Functional Dependency and Normalization :–** Integrity constraints – domain constraints, referential integrity, assertions, triggers.<br>Functional dependency, types of functional dependency, closure of set of functional dependency, canonical cover.<br>Normalization – Purpose of normalization, First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce-Codd Normal Form (BCNF), Fourth Normal Form (4NF), Fifth Normal Form (5NF), | **7 Hrs** |
| **Unit 4** | **Data Storage and Indexing :–** Overview of physical storage media, RAID, File organization, Organization of records in file, Buffer Management.<br>Indexing – Ordered indices – primary indices, secondary indices, dense and sparse indices, multilevel indexing, B tree indexing, B+ tree indexing and multiple key access.<br>Hashing – static hashing – open hashing, closed hashing, dynamic hashing.<br>Bitmap indices. | **5 Hrs** |
| **Unit 5** | **Transaction Processing and Concurrency Control :–**Transaction Processing – Concept, ACID properties, Transaction model, Schedule, Serializability – conflict and view Serializability, Recoverable schedule.<br>Concurrency Control Mechanisms – Lock based protocols, Timestamp based protocols, Thomas's Write Rule, Validation based protocols, Multi-version schemes. | **8 Hrs** |
| **Unit 6** | **Deadlock Handling and Data Recovery :–** Deadlock Handling – Deadlock prevention, deadlock detection and deadlock recovery.<br>Data Recovery – Failure Classification, Storage, Log based recovery, checkpoints, Recovery Algorithm, Buffer Management, Failure with loss of non- volatile Storage | **5 Hrs** |

| **Text Books** | |
|---|---|
| 1 | "Database System Concepts", Abraham Silberschatz, Henry F. Korth, S. Sudarshan, 6th edition, McGraw- Hill. |
| 2 | "Database Systems - A Practical Approach to Design, Implementation and Management", Thomas Connolly, Carolyn Begg, 4th Edition, Addison Wesley. |
| 3 | "MySQL Cookbook", Paul DuBois, 3rd edition, O'REILLY. |

| **Reference Books** | |
|---|---|
| 1 | "Fundamentals of Database Systems", Ramez, Elmasri, Shamkant B. Navathe, 6th Edition, Addison Wesley. |
| 2 | "Database Systems – Design, Implementation and Management", Rob & Coronel, 5th Edition, Thomson Course Technology. |

## Third Year B.Tech.
## CSL303: Machine Learning

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorial | - |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| **Total** | **100** |

| Prerequisite |
|---|
| Discrete Mathematics, Mathematics for Computer Science |

| Course Objectives<br>At the end of the course students will be able to - | |
|---|---|
| 1. | To understand Human learning aspects. |
| 2. | To understand primitives in learning process by computer. |
| 3. | To understand nature of problems solved with Machine Learning |

| Course Outcomes<br>At the end of the course students will be able to | |
|---|---|
| 1. | Explain machine learning concepts. |
| 2. | Analyse the Machine learning model. |
| 3. | Design solution using machine learning techniques. |

| | Course Contents | |
|---|---|---|
| **Unit 1.** | **Introduction to Machine Learning:**<br>Definition, Terminology, Types of learning, Machine Learning Problem categories, Machine learning architecture, process, Lifecyle, Performance measures, tools and framework, data visualization. | **6 Hrs.** |
| **Unit 2.** | **Regression:** | **8 Hrs.** |

| | | |
|---|---|---|
| | Simple regression – hypothesis, cost function, parameter learning with gradient descent, learning rate, Gradient Descent for linear regression, examples, simple regression in matrix form.<br>Multivariate Linear Regression – multiple features, hypothesis functions, Gradient Descent for multiple variables, Feature scaling, polynomial regression | |
| **Unit 3.** | **Classification- logistic regression & Neural Network:**<br>Definition, logistic regression – hypothesis representation, decision boundary, cost function, gradient descent for logistic regression. multiclass classification, Regularization - Overfitting & Underfitting, cost function, Regularized Linear Regression, Regularized Logistic Regression<br>Neural Networks- Neuron representation and model, Hypothesis for neuron, cost function, solution of a problem using single neuron. Gradient descent for a neuron.<br>Neural network, Multiclass classification with neural network.<br>Learning in neural network-backpropagation algorithm | **8 Hrs.** |
| **Unit 4.** | **Classification- Decision trees and Naïve Bayes**<br>Decision trees: definition, terminology, the need, advantages, and limitations. constructing and understanding Decision trees, common problems with Decision trees, Decision tree algorithms, random forest, examples.<br>Conditional probability and Naïve Bayes Classifier<br>Instance-based classifier – K- Nearest Neighbour Classifier | **8 Hrs.** |
| **Unit 5.** | **Unsupervised learning:**<br>Clustering, K Means clustering, Hierarchical clustering,<br>Association Rule mining | **6 Hrs.** |
| **Unit 6.** | **Recommendation System and Time series analysis**<br>Basic Text Processing with Python, regular expression, Natural Language Processing, Text Classification, Topic modeling<br>Popularity based recommender engines, Content based recommendation engines, Classification based recommendation engine, collaborative filtering<br>Date and Time Handling, Window functions, Correlation, Time Series Forecasting | **6 Hrs** |

| **Text Books** | |
|---|---|
| 1. | Machine Learning with Python- an approach to applied ML, by Abhishek Vijayvargia, BPB publications |
| 2. | Practical Machine Learning by Sunila Gollapudi Packt Publishing Ltd. |
| 3. | Machine Learning by Tom M. Mitchell, McGraw Hill Education; First edition |

| Reference Books | |
|---|---|
| 1 | Machine Learning for dummies John Paul Muller, Willey Publication |
| 2 | EthemAlpaydin : Introduction to Machine Learning, PHI 2nd Edition-2013 |

### Third Year B. Tech (Semester- V)

## CSL304 - Information Security

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | 1 Hr. /Week |
| Total Credits | 4 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Computer Network, Engineering Mathematics |

| Course Objectives | |
|---|---|
| 1 | To understand basics of cryptography, and some key encryption techniques. |
| 2 | To understand modern cryptosystems. |
| 3 | To learn security policies such as authentication, integrity and confidentiality. |
| 4 | To understand network and Web security protocols. |

| Course Outcomes<br>At the end the course students will be able to - | |
|---|---|
| 1 | Describe basic terminology in cryptography, and classical cryptosystems. |
| 2 | Explain modern cryptosystems. |
| 3 | Explain security policies such as authentication, integrity and confidentiality. |
| 4 | Explain network and Web security protocols. |

| | Contents | Hrs |
|---|---|---|
| **Unit 1.** | **Overview and Classical Encryption Techniques**<br>Overview: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security<br>Classical Encryption Techniques: Symmetric Cipher Model, Substitution Techniques, Transposition Techniques, Rotor Machines | **6 Hrs.** |
| **Unit 2.** | **Block Ciphers and Advanced Encryption Standard** | **6 Hrs.** |

| | | |
|---|---|---|
| | Block Cipher Principles, The Data Encryption Standard (DES), A DES Example, The Strength of DES, Differential and Linear Cryptanalysis, Block Cipher Design Principles<br><br>Finite Field Arithmetic, AES Structure, AES Transformation Functions, AES Key Expansion, An AES Example, AES Implementation | |
| **Unit 3.** | **Public Key Cryptography** –Principles of Public-Key Cryptosystems, The RSA Algorithm, Diffie-Hellman Key Exchange, ElGamal Cryptosystem | **5 Hrs.** |
| **Unit 4.** | **Cryptographic Data Integrity Algorithms**<br>**Cryptographic Hash Functions:** Applications of Cryptographic Hash Functions, Two Simple Hash Functions, Requirements and Security, Hash Functions Based on Cipher Block Chaining, Secure Hash Algorithm (SHA), SHA-3<br>**Message Authentication Code:** Message Authentication Requirements, Message Authentication Functions, Requirements for Message Authentication Codes, Security of MACs, MACs Based on Hash Functions: HMAC, MACs Based on Block Ciphers: DAA and CMAC, Authenticated Encryption: CCM and GCM<br>**Digital Signatures:** Elgamal Digital Signature Scheme, Schnorr Digital Signature Scheme, NIST Digital Signature Algorithm | **9 Hrs.** |
| **Unit 5.** | **Mutual Trust and Key Management**<br>**Key Management and Distribution:** Symmetric Key Distribution Using Symmetric Encryption, Symmetric Key Distribution Using Asymmetric Encryption, Distribution of Public Keys, X.509 Certificates, Public-Key Infrastructure | **5 Hrs.** |
| **Unit 6.** | **Network and Internet Security**<br>**Transport-Level Security**: Web Security Considerations, Secure Sockets Layer, Transport Layer Security, HTTPS<br>**Electronic Mail Security:** Pretty Good Privacy, S/MIME, Domain Keys Identified Mail<br>**IP Security:** IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange, Cryptographic Suites | **5 Hrs.** |

| **Text Books** | |
|---|---|
| 1 | Williams Stallings – Cryptography and Network security principles and practices, Pearson Education (LPE), Sixth Edition |

| **Reference Books** | |
|---|---|
| 1 | Cryptography and network security – Atul Kahate (TMGH) |

| | |
|---|---|
| 2 | Handbook of Applied Cryptography - Menezes, A. J., P. C. Van Oorschot, and S. A. Vanstone |

### Third Year B. Tech (Semester- V)

## CSL305 - System Programming

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | - |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Microprocessor, Computer Organization, Data Structure |

| Course Objectives | |
|---|---|
| 1 | To expose the students to fundamentals of language processing. |
| 2 | To make students to learn design of assemblers and compilers. |
| 3 | To expose the students to fundamentals of linkers, loaders & software tools for User Interfaces. |

| Course Outcomes<br>At the end of the course students will be able to - | |
|---|---|
| 1 | Explain different language processors. |
| 2 | Describe fundamentals of assemblers and macro pre-processors. |
| 3 | Explain working of compilers, linkers & loaders. |
| 4 | Summarize different software tools. |

| | Contents | |
|---|---|---|
| **Unit 1.** | **Language Processors:-** Introduction to System Software , Goals of System Software, System Programs & System Programming, Programming Languages & Language Processors, Language Processing Activities, | **7 Hrs** |

| | | |
|---|---|---|
| | Fundamentals of Language Processing, Fundamentals of Language Specification, Language Processor Development Tools. | |
| **Unit 2.** | **Assemblers:-** Elements of Assembly Language Programming, A Simple Assembly Scheme, Pass Structure of Assemblers, Design of a Two Pass Assembler, Design of single pass assembler for IBM PC 8086/8088. | **6 Hrs** |
| **Unit 3.** | **Macros and Macro Processors:-** Introduction, Macro Definition and Call, Macro Expansion, Nested Macro Calls, Advanced Macro Facilities, Design of Macro Preprocessor. | **7 Hrs** |
| **Unit 4.** | **Compilers & Interpreters :-** Causes of Large Semantic Gap, Binding & Binding Times, Scope Rules, Memory Allocation, Compilation of Expressions, Compilation of Control Structures, Code Optimization, Interpreter, Type of Interpreters. | **8 Hrs** |
| **Unit 5.** | **Linkers & Loaders :-** Introduction, Relocation & Linking Concepts, Design of a Linker, Self Relocating Programs, Linking of Overlay Structured Programs, Dynamic Linking, Loaders- Absolute Loaders, Dynamic Linking Loader, Bootstrap Loader, Relocating Loaders. | **7 Hrs** |
| **Unit 6.** | **Software Tools :-** Overview of Software Tool, Software Tools for Program Development, Editors, Debug Monitors, Programming Environments, User Interfaces | **4 Hrs** |

| **Text Books** | |
|---|---|
| 1 | Systems Programming- D.M.Dhamdhere, Mc Graw Hill Education |
| 2 | System Programming - J. J. Donovan (Mc-Graw Hill) |

### Third Year B. Tech (Semester- V)

## CSP306- Object Oriented Programming using Java

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Practical | 4 Hr. /Week |
| Total Credits | 5 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Problem analysis and program design in C, Object Oriented Design using C++ |

| Course Objectives | |
|---|---|
| 1 | To acquaint students with fundamental concepts of Java. |
| 2 | To develop problem solving ability employing object oriented programming using Java. |
| 3 | To expose students to the concepts of event handling and I/O programming features with proper exception handling. |
| 4 | To introduce students to the components of GUI development in Java and advanced framework. |

| Course Outcomes<br>At the end of the course students will be able to - | |
|---|---|
| 1 | Explain different concepts of Java. |
| 2 | Design an object oriented solution for given problem using Java. |
| 3 | Implement program using Java. |

| | Contents | |
|---|---|---|
| **Unit 1.** | **Introduction:** The Java Buzzwords, The Java Programming Environment- JVM, JIT Compiler, Byte Code Concept, HotSpot. | **7 Hrs** |

| | | |
|---|---|---|
| | A Simple Java Program, Source File Declaration Rules, Comments, Data Types, Variables, Operators, Strings, Input and Output, Control Flow, Big Numbers, Arrays- Jagged Array.<br>**Objects and Classes**: Object-Oriented Programming Concepts, Declaring Classes, Declaring Member Variables, Defining Methods, Constructor, Passing Information to a Method or a Constructor, Creating and using objects, Controlling Access to Class Members, Static Fields and Methods, this keyword, Object cloning, Class Design Hints | |
| **Unit 2.** | **Interfaces:** Defining an Interface, Implementing an Interface, Using an Interface as a Type, Evolving Interfaces, Default Methods.<br>**Inheritance:** Definition, Superclasses, and Subclasses, Overriding and hiding methods, Inheritance Hierarchies, Polymorphism, Super keyword, Final Classes and Methods, Abstract Classes and Methods, casting, Design Hints for Inheritance, Nested classes & Inner Classes, finalization and garbage collection.<br>**Packages:** Class importing, Creating a Package, Naming a Package, Using Package Members, Managing Source and Class Files. Developing and deploying (executable) Jar File. | **7 Hrs** |
| **Unit 3.** | **Exceptions:** Definition, Dealing with Errors, The Classification of Exceptions, Declaring Checked Exceptions, Throw an Exception, Creating Exception Classes, Catching Exceptions, Catching Multiple Exceptions, Re-throwing and Chaining Exceptions, finally clause, Advantages of Exceptions, Tips for Using Exceptions.<br>**I/O:** Streams, Text input and output, character streams, Reading and writing binary data in to a file. | **6 Hrs** |
| **Unit 4.** | **Introduction to the Swing**, Swing features, Creating a Frame, Positioning a Frame, Displaying Information in a Panel, The Model-View-Controller Design Pattern.<br>**Layout Management:** Introduction to Layout Management, APIs for Border Layout, panels, Grid Layout, Text Input, Choice Components, Menus, Dialog Boxes,<br>**Event Handling:** Basics of Event Handling, The AWT Event Hierarchy, Semantic and Low- Level Events in the AWT, Low-Level Event Types, Introduction to JApplet. | **6 Hrs** |
| **Unit 5.** | **Multithreading**: Processes and threads, Runnable interface, thread class, thread object, defining and starting a thread, Interrupting threads, thread states, thread properties, Joins, synchronization.<br>**Generic Programming**: Introduction, Definition of a Simple Generic Class, Generic Methods. | **7 Hrs** |
| **Unit 6.** | **Collections:** Collection interfaces, Concrete collections, The collections framework.<br>**Introduction to advanced framework in Java:** Spring, Hibernate | **6 Hrs** |

| **Text Books** | |
|---|---|
| 1 | Core Java- Volume I Fundamentals: Cay Horstmann and Gary Cornell ,Pearson, Eight edition |

| | |
|---|---|
| 2 | Core Java- Volume II Advanced Features: Cay Horstmann and Gary Cornell ,Pearson, Eight edition |

| Reference Books | |
|---|---|
| 1 | The Java Tutorials From ORACLE Java Documentation URL: http://docs.oracle.com/javase/tutorial/ (Refer For All Units) In Printed Media: The Java Tutorial: A Short Course on the Basics (6th Edition) by Raymond Gallardo, Scott Hommel, Sowmya Kannan, Publisher: Addison-Wesley Professional. |
| 2 | JAVA-The Complete Reference: Herbert Schildt, Oracle Press, Mcgraw Hill, Ninth edition |
| 3 | JAVA™ HOW TO PROGRAM, By Deitel Paul , Deitel Harvey.10th Edition, Publisher:PHI Learning |
| 4 | Core JAVA An Integrated approach: Dr.R.Nageswara Rao, Dreamtech Press. |
| 5 | A Programmer's guide to JAVA SCJP Certification: Khaleed Mughal and Rolf W. Rasmussen, Addison Wesley, Third edition |

**Practical work:** It should consist of minimum 15 experiments based on following topics. The Continuous Internal Evaluation (CIE) is based on regular practical performance and final internal practical oral examination.

| | List of Experiments |
|---|---|
| 1. | Write a program to find out day of the given date using command line argument. |
| 2. | Write a program to implement matrix operations. |
| 3. | Write a program to develop class employee with constructor to initialize instance variables. Provide Set method and Get method for instance variables. Also provide a method to raise salary of each employee by 10%. |
| 4. | Write a program to demonstrate single inheritance by creating a superclass Room and subclass Bedroom. |
| 5. | Write a program to develop class student having instance variable rn and method getno and putno. Create class Test derived from Student having instance variable as part1, part2 and method getmarks and putmarks. Define an Interface Sport having constant variable sportwt and method putwt. Derive Class Result From Test which implements this interface having data members as total. Display the result. |
| 6. | Write a program to create an area interface. Develop two different classes that implements these interface and compute area. |
| 7. | Write a program to implement mathematical package for arithmetic, statistical and trigonometric operations. |
| 8. | Write a program to develop java package for the stack and queue classes. |

| | |
|---|---|
| 9. | Write a class having two integer data members. Provide facility to add, subtract, multiply and divide these numbers. If addition goes above 1000, it generates TooLongAddition exception. If subtraction is below 0, it generates Negative Answer exception. If multiplication is above 5000, it generates TooLongMultiplication exception. |
| 10. | Write a program to remove whitespaces from a text file. Name of the file is given using command line. |
| 11. | Write a program to accept a file name from user and perform read, write/append operations on it. |
| 12. | Take Employee information such as name, employee id, department, designation, age, city, phone from user and store it in the file using DataOutputStream and FileOutputStream and Retrive data using DataInputStream and FileInputStream and display the result. |
| 13. | Write a program to develop Swing GUI based standard calculator. |
| 14. | Write a program to demonstrate key and mouse event. |
| 15. | Write a java program that implements a multi-thread application that has three threads. First thread generates random integer every 1 second and if the value is even, second thread computes the square of the number and prints. If the value is odd, the third thread will print the value of cube of the number |
| 16. | Write a program for bouncing ball application using multithreading in swing GUI. |
| 17. | Write a program to demonstrate collection and generics. |
| 18. | Write a program to create a simple Spring application. |
| 19. | Write a program to create a simple Hibernate application. |

**Third Year B.Tech.**
**CSP307: Open Source Lab I**

| Teaching Scheme | |
|---|---|
| Lectures | - |
| Practical | 2 Hrs./Week |
| Total Credits | 1 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | - |
| **Total** | **50** |

| Prerequisite |
|---|
| Open Source Technologies |

| Course Objectives | |
|---|---|
| 1. | To study free and open source software. |
| 2. | To develop applications using FOSS. |
| 3. | Apply FOSS strategy in SDLC. |

| Course Outcomes<br>At the end of the course students will be able to | |
|---|---|
| 1. | Explain benefits of Free and Open Source Software (FOSS) tools |
| 2. | Design solution to particular problem |
| 3. | Develop applications using FOSS tools |

**Course Description:**

To create an awareness among the students about free and open source software and promote the use of open source software for development of applications from various domains.

**Course Contents**

In Open Source Lab – I, the student has to study 3-4 open source software tools. It should consist of combination of applications and system software which are free and open source. The student has to perform 4 experiments based on basic python programming and 6 experiments separately on Machine learning using python or any suitable open source technology.

### Third Year B. Tech (Semester- VI)

## CSP308- Database Engineering Lab

| Teaching Scheme | |
|---|---|
| Practical | 2 Hrs. /Week |
| Total Credits | 1 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | 50 |
| Total Marks | 100 |

| Course Objectives | |
|---|---|
| 1 | To expose students to Database Management Systems functioning |
| 2 | To expose students to E-R Data Model |
| 3 | To expose students to Structured Query Language (SQL) |

| Course Outcomes<br>At the end of the course students will be able to - | |
|---|---|
| 1 | Students will be able explain the fundamentals of database management systems. |
| 2 | Students will be able to model database using E-R Diagram and design database schema. |
| 3 | Students will be able to devise SQL queries to model the reality. |

| List of Experiments - | |
|---|---|
| Minimum 8 to 10 experiments to be performed based on following guidelines | |
| 1 | Installation and Demonstration of DBMS like MySql |
| 2 | Draw E-R Diagram for different applications like – Library Management Systems, College Management Systems, Hospital Management Systems etc. |
| 3 | Convert E-R Diagrams into relational tables. |
| 4 | Use DDL Statements to Crete, Alter, Drop, Rename, Truncate Tables |
| 5 | Use DML Statements to Insert, Select, Update, Delete Data |
| 6 | Use of aggregate functions, group by – having clause and order by clause. |
| 7 | Use of Joins |
| 8 | Use of Set Operations |
| 9 | Creation of Indices and Views in SQL |
| 10 | Find Canonical Cover and Closure for set of functional dependencies. |
| 11 | Demonstration of Indexing – Dense index, Sparse index, B+ tree index |

| | |
|---|---|
| 12 | Demonstration of Hashing – Static hashing, Dynamic hashing |
| 13 | Demonstration of Log based recovery. |
| 14 | Study of concurrency control mechanisms |

## Third Year B.Tech.
## CSS309: Seminar

| Teaching Scheme | |
|---|---|
| Tutorial | 1 Hrs./Week |
| Total Credits | 1 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | - |
| **Total** | **50** |

| CO NO | CO Statements |
|---|---|
| **CO1** | **Student will be able to select a technical topic** |
| **CO2** | **Student will be able to give presentation in team on selected technical topic** |
| **CO3** | **Student will be able to write a report on presented technical topic** |

A group of 3 to 5 students is required to do a seminar presentation on a latest and recent topic in computer science. This topic should preferably be the foundation of the project students wish to work for their final year. Preparation and presentation of a seminar is intended to investigate an in-depth review of literature, prepare a critical review and develop confidence to present the material by the students. The seminar shall be evaluated by a Department Committee constituted for this purpose, based on a report submitted by the candidate and a viva-voce conducted at the end of the semester.

**Third Year B.Tech.**
**CSI310: Softskills**

| Teaching Scheme | |
|---|---|
| Lecture | 2 Hrs./Week |

| Evaluation Scheme | |
|---|---|
| **Total** | **Grade** |

| Course Objectives<br>At the end of the course students will be able to - | |
|---|---|
| 1. | Enhance the communications skills of the students. |
| 2. | Expose the students to basic skills of team work |
| 3. | Inculcate the writing skills necessary for business communications. |

| CO NO | CO Statements |
|---|---|
| CO1 | Students Should be able to exhibit spoken, writing and listening skills effectively. |
| CO2 | Students should be able to participate in group discussion, debates, and interviews to exhibit the team building skills |
| CO3 | students Should be able to exhibit effective presentation skills. |

| | Course Contents |
|---|---|
| **Unit 1.** | **Communication Skills Verbal Communication**<br>Effective Communication - Active listening – Articulation Paraphrasing – Feedback Non Verbal Communication - Body Language of self and others Importance of feelings in communication - dealing with feelings in communication Inter and Intrapersonal communication- Self-esteem and confidence - Assertiveness |
| **Unit 2.** | **Importance of Team work Self Enhancement**<br>Importance of developing assertive skills- developing self-confidence – developing emotional intelligence. Importance of Team work – Team vs. Group - |

| | |
|---|---|
| | Attributes of a successful team – Barriers involved Working with Groups – Dealing with People- Group Decision Making Effective teams – Elements of Team work - Stages in team formation |
| **Unit 3.** | **Writing**<br>Introduction to writing, Hallmark of good writing, Writing conventions, business writing, writing a notice, writing styles, e-mail writing, report writing, practice. |

| **Text Books** | |
|---|---|
| 1 | Infosys Softskills Manual |
| 2 | Developing Communication Skills by Krishna Mohan and Meera Banerji; MacMillan India Ltd., Delhi Eight edition (Unit 3 I/O and Unit 6 Database programming) |
| 3 | Essentials of Effective Communication, Ludlow and Panthon; Prentice Hall of India. |
| 4 | Good To Great - Jim Collins |

# Third Year B. Tech. Computer Science and Engineering – Semester – VI

## Third Year B.Tech.
## CSL311: OPERATING SYSTEM-II

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorial | - |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| **Total** | **100** |

| Prerequisite |
|---|
| Operating System – I, Data Structures and Algorithms |

| Course Objectives | |
|---|---|
| 1. | To provide knowledge to the students about Fundamental architecture of UNIX operating system kernel and algorithms of buffer cache management. |
| 2. | To describe Internal File system organizations and related algorithms in UNIX |
| 3. | To provide knowledge to the students about System calls for UNIX file system, Process structure, creation and management in UNIX. |
| 4. | To provide knowledge to the students about Architecture and algorithms of process scheduling and I/O subsystem as well as memory management. |

| Course Outcomes |
|---|
| At the end of the course students will be able to |

| | |
|---|---|
| 1. | Explain structure and working of various subsystems in UNIX Kernel |
| 2. | Describe algorithms used in various subsystems of UNIX Kernel. |
| 3. | Analyze algorithms and concepts used in UNIX kernel |

| | Course Contents | |
|---|---|---|
| **Unit 1.** | **Introduction and buffer cache:-** General Overview of the System - History, System Structure, User Perspective, Operating System Services, Assumption About Hardware, Architecture of UNIX OS, Introduction to system concepts, Kernel Data Structure, System Administration,<br>**Buffer Cache: -** Buffer headers, structure of the buffer pool, scenarios for retrieval of a buffer, reading and writing disk blocks, advantages and disadvantages of cache. | **7 Hrs.** |
| **Unit 2.** | **Internal Representation of Files:-**I-nodes, structure of the regular file, directories, conversion of a pathname to i-node, super block, I-node assignment to a new file, allocation of disk blocks, other file types. | **6 Hrs.** |
| **Unit 3.** | **System Calls for file system:-** System Calls for file system:- Open, Read, write, File and Record Locking, Adjusting the position of FILE I/O-LSEEK, Close, File Creation, Creation of Special File, Change Directory and Change Root, Change Owner and Change Mode, Stat and fstat, Pipes, Dup, Mounting and Un-mounting file systems, Link, Unlink, File System Abstractions, File system maintenance | **6 Hrs.** |
| **Unit 4.** | **The Structure of process:-** Process stages and transitions, layout of system memory, the context of a process, Saving context of a process, manipulation of the process address space | **6 Hrs.** |
| **Unit 5.** | **Process Control and Scheduling:-**Process Control: - Process creation, signals, process termination, awaiting process termination, invoking other programs, the user id of a process, the shell, System Boot and the Init process.<br>**Process Scheduling**: - Process Scheduling, system call for time, clock | **6 Hrs.** |
| **Unit 6.** | **Memory management and I/O Subsystem:-**Swapping, Demand passing, a hybrid system with demand paging and swapping. Driver interfaces, disk drives, terminal drivers, Streams. | **7 Hrs** |

| Text Books | |
|---|---|
| 1. | The design of Unix Operating System - Maurice J. Bach (PHI) |

| Reference Books | |
|---|---|
| 1 | Linux System Programming - Robert Love, Publisher - SPD, O' REILLY |
| 2 | Unix concepts and administration – 3rd Edition – Sumitabha Das (TMGH). |

### Third Year B. Tech (Semester- VI)

## CSL312- Compiler Construction

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | 1 Hrs. / Week |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| System Programming, Automata Theory |

| Course Objectives | |
|---|---|
| 1 | To introduce the fundamentals of compilers and their phases. |
| 2 | To design and implement phases of compiler. |
| 3 | To learn various open source tools like Lex and YACC. |

| Course Outcomes<br>At the end of the course students will be able to - | |
|---|---|
| 1 | Explain concepts related to the phases of compiler. |
| 2 | Design the parser and its related parsing tables. |
| 3 | Describe the syntax directed translation & intermediate code generation. |
| 4 | Explain various techniques of code optimization and code generation. |

| | Contents | |
|---|---|---|
| **Unit 1.** | **Introduction to Compiling:-** Compilers, Phases of a compiler, Compiler construction tools, cousins of the compiler | **04 Hrs.** |

| | | |
|---|---|---|
| **Unit 2.** | **Lexical Analysis:-** Role of a Lexical analyzer, input buffering, specification and recognition of tokens, finite automata implications, designing a lexical analyzer generator. | **07 Hrs.** |
| **Unit 3.** | **Syntax Analysis:-**Role of Parser, Writing grammars for context free environments, Top-down parsing, Recursive descent and predictive parsers (LL), Bottom-Up parsing, Operator precedence parsing, LR, SLR and LALR parsers. | **08 Hrs.** |
| **Unit 4.** | **Syntax Directed Translation and Intermediate Code Generation:-** Syntax directed definitions, construction of syntax tree, S-attributed definitions, L-attributed definitions, Intermediate languages, assignment statements, back patching, procedure calls | **07 Hrs.** |
| **Unit 5.** | **Run Time Environments :-** Source language issues, storage organization and allocation strategies, parameter passing, symbol table organizations and generations, dynamic storage allocations. | **05 Hrs.** |
| **Unit 6.** | **Code Optimization & Code Generation:-** Sources of optimization, Peephole optimization and basic blocks, loops in flow graphs, Data flow analysis and equations, Issues in design of a code generator and target machine, Basic blocks and flow graphs, Issues of register allocation, code generation from Dags. | **08 Hrs.** |

| **Text Books** | |
|---|---|
| 1 | Compilers - Principles, Techniques and Tools - A.V. Aho, R. Shethi and J.D. Ullman (Pearson Education.) 3rd Edition |

| **Reference Books** | |
|---|---|
| 1 | Crafting A Compiler with C - Charles Fischer, Richard LeBlanc (Pearson publication) (For practical use only) |
| 2 | Modern Compiler Design - D. Grune, H. Bal, C. Jacobs, K. Langendoen (Wiley publication) (For practical use only). |

### Third Year B. Tech (Semester- VI)

## CSL313- Computer Algorithm

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | 1 Hr. /Week |
| Total Credits | 4 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Problem analysis and program design in C, Object Oriented Design using C++ , Data Structure |

| Course Objectives | |
|---|---|
| 1 | To expose students to fundamentals of algorithms. |
| 2 | To expose students to different algorithm design paradigms. |
| 3 | To expose students to performance analysis of algorithms. |
| 4 | To expose students to different data structures and operations on them. |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | Explain fundamentals of algorithms. |
| 2 | Apply Divide and Conquer and Greedy Approach to tackle problems. |
| 3 | Solve the problems using dynamic programming and backtracking. |
| 4 | Analyse performance of algorithms using asymptotic analysis. |
| 5 | List different data structures and operations on them. |

| | Contents | |
|---|---|---|
| **Unit 1.** | **Introduction, Divide and Conquer** – Introduction, Characteristics of algorithms, Pseudo code Conventions, Recursive Algorithms, Performance | **8 Hrs.** |

| | | |
|---|---|---|
| | analysis, Asymptotic notations (O, Ω, Θ), Performance measurement, Randomized Algorithms, Recurrence relations.<br>**Divide and Conquer**– General method, Binary search, finding the maximum and minimum, Merge sort, Quick sort, Selection, Convex Hull. | |
| **Unit 2.** | **The Greedy method** – General method, , Knapsack problem, Job sequencing with deadlines, minimum-cost spanning trees – Prim's and Kruskal's Algorithms, Optimal storage on tapes , Optimal merge patterns, Huffman codes Single source shortest paths. | **7 Hrs.** |
| **Unit 3.** | **Dynamic Programming** – General method, Multistage graphs, All pair shortest paths, Optimal binary search trees, 0/1 knapsack, Reliability design, Traveling Sales person problem, Flow shop scheduling. | **7 Hrs.** |
| **Unit 4.** | **Backtracking –** General method, n-queen problem, Permutation tree, Sum of subsets, Knapsack Problem, Hamiltonian Cycle, and Graph Coloring. | **5 Hrs.** |
| **Unit 5.** | **NP Hard and NP Complete Problems –** Basic Concepts, P, NP, NP Complete, NP Hard, Cook's Theorem, NP Hard Graph problems, NP Hard Scheduling problems, NP Hard Code Generation Problems. | **6 Hrs.** |
| **Unit 6.** | **Approximation Algorithms –** Introduction, Absolute approximations, ε-approximations, Polynomial time approximation Schemes, Fully Polynomial Time Approximation Schemes, Probabilistically Good Algorithms. | **6 Hrs.** |

| **Text Books** | |
|---|---|
| 1 | "Fundamentals of Computer Algorithms", Ellis Horowitz, SartajSahni, SanguthevarRajasekaran, 2nd Edition, Universities Press. |
| 2 | "Introduction to Algorithms", Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein,<br>3rd Edition, The MIT Press. |

| **Reference Books** | |
|---|---|
| 1 | "The Design and Analysis of Computer Algorithms", A. Aho, J. Hopcroft and J. Ullman, 1st Edition, Addison-Wesley |
| 2 | "Introduction to The Design and Analysis of Algorithms", Anany Levitin, 3rd Edition, Pearson. |

**Third Year B. Tech (Semester- VI)**

## CSL314 – Data Science

| Teaching Scheme | |
|---|---|
| Lectures | 03 Hrs. /Week |
| Tutorials | -- |
| Total Credits | 03 |

| Evaluation Scheme | |
|---|---|
| SE-I | **25** |
| SE-II | **25** |
| SEE | **50** |
| Total | 100 |

| Prerequisite |
|---|
| Machine Learning, Statistics, Database |

| Course Objectives | |
|---|---|
| 1 | To expose the students to fundamentals of data science. |
| 2 | To make students to learn techniques of data processing and data analytics. |
| 3 | To make students to learn how to extract meaning from data. |
| 4 | To understand Data Visualization and Social Network Analysis techniques. |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | Explain fundamentals of data science. |
| 2 | Apply data processing and exploratory data analytics. |
| 3 | Explain fundamental concepts of unstructured data mining. |
| 4 | Explain Social Network Analysis techniques |
| 5 | Explain model evaluation techniques. |

| | Contents | |
|---|---|---|

| | | |
|---|---|---|
| **Unit 1.** | **Introduction**<br>Introduction to Data Science, Examples, Data Sources, Challenges, Applications, Comparative Study of data science with databases, scientific computing, computational science, machine learning, Data Modeling, Statistical Data Modeling, Computational Data Modeling, Statistical limits on data- Bonferroni's principle. | **06 Hrs** |
| **Unit 2.** | **Data Preprocessing**<br>Why do We Need to Preprocess the Data?, Data Cleaning, Handling Missing Data, Identifying Misclassifications, Graphical Methods for Identifying Outliers, Measures of Center and Spread, Data Transformation, Min–Max Normalization, Z-Score Standardization, Decimal Scaling, Transformations to Achieve Normality, Numerical Methods for Identifying Outliers, Flag Variables, Transforming Categorical Variables into Numerical Variables, Binning Numerical Variables, Reclassifying Categorical Variables, Adding an Index Field, Removing Variables that are not Useful, Variables that Should Probably not be Removed, Removal of Duplicate Records | **07 Hrs** |
| **Unit 3.** | **Exploratory Data Analysis**<br>Hypothesis Testing Versus Exploratory Data Analysis, Getting to Know the Data Set, Exploring Categorical Variables, Exploring Numeric Variables, Exploring Multivariate Relationships, Selecting Interesting Subsets of the Data for Further Investigation, Using EDA to Uncover Anomalous Fields, Binning Based on Predictive Value, Deriving New Variables: Flag Variables, Deriving New Variables: Numerical Variables, Using EDA to Investigate Correlated Predictor Variables | **07 Hrs** |
| **Unit 4.** | **Extracting Meaning from Data**<br>William Cukierski, The Kaggle Model, Thought Experiment: What Are the Ethical Implications of a Robo-Grader?, Feature Selection, David Huffaker: Google's Hybrid Approach to Social Research | **06 Hrs** |
| **Unit 5.** | **Data Visualization and Fraud Detection**<br>Data Visualization History, What Is Data Science, Redux?, A Sample of Data Visualization Projects, Mark's Data Visualization Projects, Data Science and Risk, Data Visualization at Square, Ian's Thought Experiment, Data Visualization for the Rest of Us | **06 Hrs** |
| **Unit 6.** | **Social Networks and Data Journalism**<br>Social Network Analysis at Morning Analytics, Social Network Analysis, Terminology from Social Networks, Thought Experiment, Morningside Analytics, More Background on Social Network Analysis from a Statistical Point of View, Data Journalism, | **06 Hrs** |

| **Text Books** | |
|---|---|
| 1 | Cathy O'Neil and Rachel Schutt, "Doing Data Science", O'Reilly Media, October 2013, Print ISBN:978-1-449-35865-5 |
| 2 | Daniel T. Larose and Chantal D. Larose, "Data Mining and Predictive Analytics", Wiley Publication, ISBN 978-1-118-11619-7 |

| Reference Books | |
|---|---|
| 1 | Field Cady, "The Data Science Handbook", Wiley Publication |
| 2 | Wes McKinney, "Python for Data Analysis", O'Reilly Media, ISBN ISBN: 978-1-449-31979-3 |
| 3 | Jure Leskovec, Anand Rajaraman, and Jeffery David Ullman, "Mining of Massive Datasets", Cambridge University Press, 2 edition (13 November 2014), ISBN-10: 1107077230, ISBN-13: 978-1107077232 |

### Third Year B. Tech (Semester- VI)

## ETLOE1 – Fundamentals of Embedded Systems

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | - |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Microprocessors, Computer Organizations |

| Course Objectives | |
|---|---|
| 1 | To learn fundamentals components of Embedded Systems. |
| 2 | To understand and gain knowledge of architecture of 8051 microcontroller. |
| 3 | To understand Arduino hardware and programming IDE. |
| 4 | Explore the provided example code and online resources for extending knowledge about the capabilities of the Arduino microcontroller |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | **Compare and Contrast** basic architecture of 8051 Microcontrollers and Arduino UNO. (Understand) |
| 2 | **Describe** various on chip resources like I/O ports, Timer, Serial Communication and Interrupt structure of 8051 Microcontroller and Arduino UNO (Understand). |

| | |
|---|---|
| 3 | **Demonstrate** PORT programming skills for interfacing external hardware's with 8051 Microcontroller and Arduino UNO (Apply). |
| 4 | **Compose** well-structured and understandable programs for Microcontroller 8051 and Arduino UNO (Apply) |

| Course Contents | | |
|---|---|---|
| **Unit 1.** | **Introduction to MCS51 Microcontroller Family:-**Introduction to MCS51Family, microcontroller and embedded processors, Architecture, Functional Pin out diagram, Programming Model, Memory Organization, data types, directives, comparison between different MCS51 families, Reset Circuit, Machine Cycle, Oscillator Circuit. | **7 Hrs** |
| **Unit 2.** | **8051 ON Chip Resources, Programming and Interfacing:-**Input / Output Ports, Counters & Timers, Serial Communication, Interrupt programming [Structure of Above, Related S.F.R, Instruction, Associated Programs] | **7 Hrs** |
| **Unit 3.** | **Introduction to Embedded C programming for 8051:-**Data types and time delay, I/O programming, Logic operations, data conversion, accessing code ROM space, data serialization in 8051 using embedded C programming. | **8 Hrs.** |
| **Unit 4.** | **Hardware Interfacing & Application:-**Interfacing: LED, Switch, Relay, LCD, Analog to Digital Converter, Digital to Analog Converter, L293D motor driver IC, Serial Communication with RS 232.<br>[Note: Above hardware interfacing can be done by using Embedded C programming model] | **7 Hrs.** |
| **Unit 5.** | **Introduction to Arduino Platform:-**The Arduino Platform, Block diagram, Architecture, Pin functions, Different types of Arduino board Overview of main features such as I/O Ports, Timers, interrupts serial port, PWM, ADC etc. Introduction to Arduino IDE, writing, saving, compiling and uploading sketches. | **6 Hrs** |
| **Unit 6.** | **Arduino Board Programming:-**Arduino Control structure, Functions, operators, sketch structure, Interfacing different hardware's- LED, Liquid Crystal Display[LCD], Sensors, Motor Driver IC-L293D, Ethernet shield. | **7 Hrs** |

| Reference Books | |
|---|---|
| 1 | The 8051 Microcontroller & Embedded Systems using Assembly and C By Muhammad Ali Mazidi, Janice GillispieMazidi&RolinD.McKinlay, Pearson Edition L. P .E. |

| | |
|---|---|
| 2 | The 8051 Microcontroller By Ayala 3rdEdition |
| 3 | 8051 Microcontroller-Internals ,Instructions, Programming & Interfacing by SubrataGhoshal,,Pearson |
| 4 | Programming Arduino Getting Started with Sketches By Simon Monk, McGrawHill |
| 5 | Beginning Arduino Programming By Brian Evans, Technology in Action. |

**Third Year B. Tech (Semester- VI)**

# MELOE2 – Mechatronics

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | - |
| Total Credits | 3 |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Basic Mechanical Engineering, Basic Electrical Engineering |

| Course Objectives | |
|---|---|
| 1 | To introduce students to various concepts of automation, Mechatronics and PLC and the integration of different branches of engineering in Mechatronics. |
| 2 | To make students aware of the recent trends and practices in Mechatronics in manufacturing and service sector for productivity improvement and cost, time and human intervention reduction. |
| 3 | To analyze the real life industrial problems in traditional systems and convert them into highly effective Mechatronics systems. |
| 4 | To design, analyze, modify, if required, and implement ladder programs using PLC for various industrial, home and office automation problems along with necessary fault finding. |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | Describe and discuss sensors, PLC, digital circuits and signal conditioning. |
| 2 | Distinguish between traditional and mechatronics system and justify the given system is mechatronics system. |
| 3 | Design solutions for industrial automation problems using timers, counters, internal relays and other programming instructions. |
| 4 | Construct and communicate automated solutions for economic, relevant applications in respective industries and environmental problems. |

| | Contents | |
|---|---|---|
| **Unit 1.** | **Introduction to Mechatronics:-** Introduction to Mechatronics: What is Mechatronics, Mechatronics systems, Measurement systems, Control systems, Multidisciplinary scenario. Traditional Vs Mechatronics Design, Case studies of mechatronics system designs like piece counting system, | **05 Hrs** |

| | | |
|---|---|---|
| | pick and place manipulator, part loading, unloading and handling system and similar systems. | |
| **Unit 2.** | **Transducers & Sensors:-**Performance Terminology, Position Sensors: Limit switch, photoelectric switches, proximity sensors, pneumatic limit valves and backpressure sensors, pressure switches, resolvers, incremental & absolute encoders, decoders & relays. Displacement: Potentiometer sensors, LVDT, capacitive displacement sensors. Velocity sensors: Tachogenerator, use of encoders, Temperature sensors, Selection of sensors. | **07 Hrs** |
| **Unit 3.** | **Signal Conditioning:-** Signal conditioning process, Operational amplifier (inverting amplifier, non-inverting amplifier, summing, integrating amplifier), protection, filtering, data acquisition, multiplexer, analog to digital converter (ADC), digital to analog converter (DAC). Sample and hold, Interfacing input output ports, interfacing requirements, buffer, handshaking, polling and interrupts. | **06 Hrs** |
| **Unit 4.** | **Introduction to Fluid Power:-** Introduction to Hydraulic and Pneumatic Systems. Basic Components. Symbols - (Including fundamentals of fluid flow, fluids etc.), Hydraulic pumps/motors/actuators, Development of hydraulic circuit, Basic design, Meter In and Meter Out Circuits, Basic Pneumatic system and components, Pneumatic valves, Pneumatic Circuits & Systems, Fluid Logic, Application of Hydraulics & Pneumatics in industrial Automation. | **07 Hrs** |
| **Unit 5.** | **Programmable Logic Controller (PLC):-** Introduction, Definition of PLC, PLC system and components of PLC input output module, PLC advantages and disadvantages, RS-232 serial interface, Block diagram for interfacing of PLC, computer and system to be controlled, Machine control terminology, update – solve ladder – update, physical components Vs. program components, light control example, disagreement circuit, majority circuit, oscillator, holding (sealed or latches) contacts, always ON always OFF contacts, fail safe circuit, AND-OR and OR-AND circuits. Introduction to Supervisory Control and Data Acquisition System (SCADA) and applications. | **07 Hrs** |
| **Unit 6.** | **Ladder diagram & PLC programming fundamentals:-**Basic components and other symbols, Fundamentals of ladder diagram, PLC input instructions, outputs, coils, indicators, operational procedures, contact and coil input output, programming example, simple industrial applications, Nesting of ladders. PLC timer functions – Introduction, timer functions, industrial applications, industrial process timing applications, PLC control functions – PLC counters and its industrial applications, Internal Relays, PLC system fault finding – Program testing, | **07 Hrs** |

| | | |
|---|---|---|
| | testing inputs and outputs, Fault detection techniques, common hardware faults. | |

| Text Books | |
|---|---|
| 1 | Mechatronics – W. Bolton, Pearson education |
| 2 | Mechatronics – Mahalik, TATA McGraw Hill |
| 3 | Programmable logical controller, Reis Webb, Prentice Hall |

| Reference Books | |
|---|---|
| 1 | Microprocessor 8085 – Gaonkar |
| 2 | Mechatronics – Appu Kuttam, Oxford publications |
| 3 | Automated Manufacturing systems, S. Brain Morris, McGRaw Hill |
| 4 | Introduction to PLC programming, NIIT, P. |
| 5 | Programmable logical controller, Hackworth & Hackworth, Pearson Education |
| 6 | Programmable logical controller,3e Gary Dunning Cengage Learning |

**Third Year B. Tech (Semester- VI)**

## TFLOE1 – Merchandising

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Tutorials | - |
| Total Credits | 3 |
| | |

| Evaluation Scheme | |
|---|---|
| SE-I | 25 |
| SE-II | 25 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| - |

| Course Objectives | |
|---|---|
| 1 | To explain the organization of the business |
| 2 | To describe the marketing and merchandising process. |
| 3 | To describe retailing and Global Sourcing Strategies, Supply Chain and demand |
| 4 | To distinguish between principle and auxiliary documents. |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | Explain the organization of the business |
| 2 | Describe the marketing and merchandising process. |
| 3 | Describe retailing and Global Sourcing Strategies, Supply Chain and demand chain analysis |
| 4 | Distinguish between principle and auxiliary documents. |

| Course Contents | | |
|---|---|---|
| **Unit 1.** | **Organization of the Business:-**Introduction to industry - organization structure of an industry Business, different types of organization structure, supply chain of business, different departments of an organization and their function. Standard operating procedure and their importance | **6 Hrs.** |
| **Unit 2.** | **Marketing:-**Marketing, different definitions of marketing, Marketing evolution, Product life cycle, selling vs marketing, marketing environment, marketing research, marketing objectives and Strategies, marketing mix, | **7 Hrs.** |

| | | |
|---|---|---|
| | marketing planning, Market sourcing- domestic, export manufacturing, retailers/wholesalers/co-operative, Buying agencies/Offices, Direct exporting, different case studies. | |
| **Unit 3.** | **Product development:-** Introduction, Process of product development, Different types of samples, sample approval procedure, sample review, merchandiser's role in product development, Bill of material. | **6 Hrs.** |
| **Unit 4.** | **Merchandising:-**Introduction to merchandising and its process, roles and responsibilities of merchandiser in different organizations, categories of apparel merchandising, buying cycles and tools of merchandising – buying cycle, time and action plan, range planning, critical path, Costing techniques and Spec Sheets. Visual Merchandising | **7 Hrs.** |
| **Unit 5.** | **Sourcing:-**Need for sourcing, Resource Planning – Global Sourcing, Strategies, Supply Chain and demand chain analysis, Supply chain management and its importance. JIT technology. Buying house –Its function and role in industry. | **6 hrs.** |
| **Unit 6.** | **Export Documentation:-**Various types of export documents, Pre-shipment Post -shipment documentation, Terms of sale, payment, shipment etc. Different Inco terms, Export incentives: Duty drawback, DEPB, I / E license - exchange control regulation – (FEMA) foreign exchange management acts - export management risk - export finance. WTO / GATT / MFA - Functions and objectives, successes and failures. | **7 Hrs.** |

| Reference Books | |
|---|---|
| 1 | Marketing Management by Philip Kotler. 15th edition Pearson Education. ISBN: 978-9332557185 |
| 2 | Cooklin's Garment Technology for Fashion Designers, 2nd Edition by Gerry Cooklin, Steven Hayes, John McLoughlin, Dorothy Fairclough, Blackwell Publications, ISBN: 978-1-4051-9974-2 |
| 3 | Garment Manufacturing: Processes, Practices and Technology by Prasanta Sarkar, Online Clothing Study. ISBN: 978-9383701759 |
| 4 | Fashion Buying by Elaine Stone. McGraw-Hill In publication ISBN: 978-0070617469 |
| 5 | Apparel Merchandising by Kumar, Abhishek Publications, ISBN: 978-8182473010 |
| 6 | Fashion Marketing by Mike Easey. john Wiley & Sons publication. ISBN: 978-0632034598 |
| 7 | Apparel manufacturing hand book by Jacob solinger, bobbin media corp publication. |

**Third Year B. Tech (Semester- VI)**

## CSP315- Web Technologies

| Teaching Scheme | |
|---|---|
| Lectures | 3 Hrs. /Week |
| Practical | 4 Hrs. /Week |
| Total Credits | 5 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | 50 |
| Total | 100 |

| Prerequisite |
|---|
| Computer Network. |

| Course Objectives | |
|---|---|
| 1 | To expose students to emerging web technologies and tools |
| 2 | To introduce client and server side technologies required for web applications development |
| 3 | To know XML concepts and its applications |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | Describe different concepts of web technology |
| 2 | Develop an application to handle XML document |
| 3 | Develop web application using client side technology. |
| 4 | Develop web application using server side technology. |

| | Contents | |
|---|---|---|
| **Unit 1.** | **HTML Design Patterns:** HTML Structure, Terminal Block Elements, Structural Block Elements, Multipurpose Block Elements, Inline Elements, Class and ID Attributes, HTML Whitespaces<br>**CSS Selector and Inheritance:** Type, Class and ID Selector, Position and Group Selectors, Attribute Selectors, Pseudo-element Selectors, Pseudo-class Selectors, Subclass Selector, Inheritance, Visual Inheritance<br>**Box Model:** Display, Box Model, Inline Box, Inline-Block Box, Block Box, Table Box, Absolute Box, Floated Box. | **06 Hrs** |
| **Unit 2.** | **Responsive Web Designing:** Introduction, Viewport, Grid View, Image, Video Media Queries, RWD frameworks<br>**Twitter Bootstrap :** Introduction, Grid Basics, Typography, Tables, Images, Jumbotron, Well, Alerts, Button, Button Group, Glyphicons, Borders, Labels, Progress bar, Pagination, Pager, List groups, Panels, Drowpdown, Collapse, Tabs, Navbar, Forms, Inputs, Input sizing, Media Objects, Carousel, Modal, Tooltip, Popover, Scrollspy, Affix, Filters. | **06 Hrs** |

| | | |
|---|---|---|
| **Unit 3.** | **XML and Parsing:** What is XML, XML verses HTML, XML terminology, XML standards, XML syntax checking, The idea of markup, XML Structure, Organizing information in XML, Creating Well-formed XML, XML Namespaces. DTD- Introduction to DTD, Document Type Declaration, Element Type Declaration, Attribute Declaration, Conditional Section, Limitations of DTD<br>Introduction to Parser, Parsing approaches, JAXP, JAXP and SAX, JAXP and DOM.<br>Introduction to XSL, overview, XPATH, XSLT – templates, creating elements and attributes, looping and sorting, conditional processing, defining variables. | **06 Hrs** |
| **Unit 4.** | **JavaScript:** Introduction, Core features - Data types and Variables, Operators, Expressions and Statements, Functions & Scope, Objects - Array, Date and Math related Objects, Document Object Model, Event Handling, Browser Object Model, Windows and Documents, Form handling and validations. | **06 Hrs** |
| **Unit 5.** | **jQuery:** Introducing jQuery, jQuery selector, jQuery HTML, Animation effects, Event handling, DOM, jQuery DOM traversing, DOM manipulation. | **06 Hrs** |
| **Unit 6.** | **Introducing PHP:** History, General Language Feature<br>**PHP Basics:** Embedding PHP code in Your Web Pages, Commenting Your Code, Outputting Data to the Browser, PHP supported Data Types, Identifiers, Variables, Constants, Expressions, String Interpolation, and Control Structures<br>**Functions:** Invoking a Function, Creating a Function, Function Libraries<br>**Array:** Introduction, Creating an array, outputting an Array, Merging, slicing, splicing and Dissecting Arrays, Other useful Array Functions<br>**Using PHP with MySQL:** Installation Prerequisites, Using the MySqli Extension, Interacting with the Database, Executing Database Transactions<br>**Session Handlers:** What Is Session Handling, Configuration Directives, Working with Sessions, Practical Session-Handling Examples, Creating Custom Session Handlers<br>**Handling File Uploads:** Uploading Files with PHP | **08 Hrs** |

| **Text Books** | |
|---|---|
| 1 | Pro HTML5 and CSS3 Design Patterns by Michael Bowers, Dionysios Synodinos and Victor Sumner, Apress edition |
| 2 | Twitter Bootstrap Development How to by David Cochran, Packt Publication |
| 3 | XML and Related Technologies – Atul Kahate , Pearson Education. |
| 4 | JavaScript: The Definitive Guide by David Flanagan, O'Reilly Media |
| 5 | jQuery in Action by Bear Bibeault, Manning Publication |
| 6 | Beginning PHP and MySQL: From Novice to Professional, Fourth Edition - W. Jason Gilmore |

| Reference Books | |
|---|---|
| 1 | Beginning with HTML5 and CSS3 The Web Evolved by Murphy, Apress |
| 2 | Responsive Web Design with HTML5 and CSS3 by Ben Frain, Packt Publication |
| 3 | JavaScript: The Complete Reference by Thomas A Powell, Fritz Schneider, Tata McGraw Hill |
| 4 | Head First jQuery by Ryan Benedetti, O'reilly Publication |
| 5 | Modern PHP by Josh Lockhart, O'reilly Publication |

**Practical Work:** It should consist of minimum 15 experiments based on following topics. The Continuous Internal Evaluation (CIE) is based on regular practical performance and final internal practical oral examination.

1. Developing web pages with different layout
2. Applying CSS to developed pages
3. Creating responsive web pages with Bootstrap
4. Validating forms using JavaScript
5. Event handling with JavaScript
6. Performing processing and providing alerts with JavaScript on form data.
7. Developing animation effects with jQuery
8. Event handling with jQuery
9. Traversing DOM tree with jQuery
10. DOM manipulation with jQuery
11. Program based on PHP variables, Expression, arrays, control structure
12. Upload various types of file from client side to server with validation
13. Insert user entered data in form to MySQL database using PHP
14. Update user's data stored in MySQL database using PHP
15. Managing sessions in PHP having login facility in any web application

## Third Year B.Tech.
## CSP316: Open Source Lab II

| Teaching Scheme | |
|---|---|
| Practical | 2 Hrs/Week |
| Total Credits | 1 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | - |
| **Total** | **50** |

| Course Objectives | |
|---|---|
| 1. | To study free and open source software. |
| 2. | To develop applications using FOSS. |
| 3. | Apply FOSS strategy in SDLC. |

| Course Outcomes<br>At the end of the course students will be able to | |
|---|---|
| 1. | Elaborate use of FOSS tools. |
| 2. | Design solution to particular problem using FOSS. |
| 3. | Inculcate FOSS strategy in SDLC to design and develop applications. |

### Course Description:

To create an awareness among the students about free and open source software and promote the use of open source software for development of applications from various domains.

### Course Contents

In Open Source Lab – I, the student has to study 3-4 open source software tools. It should consist of combination of applications and system software which are free and open source. The student has to perform 6 experiments in Data Science using Python, Hadoop, Sparc, Hive etc. and 4 experiments separately on open source operating systems like Linux.

### Third Year B. Tech (Semester- VI)

## CSD317- Mini project II

| Teaching Scheme | |
|---|---|
| Practical | 2 Hrs. /Week |
| Tutorials | - |
| Total Credits | 2 |

| Evaluation Scheme | |
|---|---|
| CIE | 50 |
| SEE | 50 |
| Total | 100 |

| Course Objectives | |
|---|---|
| 1 | To expose the students to use engineering approach to solve domain specific real time problem. |
| 2 | To use the appropriate and newer technologies while developing the project. |
| 3 | To learn the skills of team building and team work. |

| Course Outcomes<br>At the end of the course students will be able to- | |
|---|---|
| 1 | Analyse the problem and prepare SRS and design document. |
| 2 | Write code |
| 3 | Carry out testing. |
| 4 | Write a report covering details of the project and give presentation on a project. |

### Contents

The students should form group of maximum 5 students and every group is supposed to choose a specific domain to do the mini project-II. Further the group should identify the relevant problem in the selected domain and propose the solution, which can be implemented as a mini-project using suitable technology. The mini-project-II work should be evaluated by a team of teachers appointed by the department/COE. The evaluation and marking should include Continuous Internal Evaluation (CIE) and Semester End Examination (SEE) during which the group should give presentation and demonstration of their work done. Care should be taken to avoid out-sourcing of the work.

**Third Year B. Tech (Semester- VI)**

## CST318- Industrial Training / Internship

| Course Objectives | |
|---|---|
| 1 | To expose the students to actual working environment in industry to enhance their employability skills and become job ready |
| 2 | To enhance their knowledge and skill from what they have learned in the syllabus |
| 3 | To enhance students knowledge in a particular technology |
| 4 | To enhance student's leadership ability and responsibility to perform or execute the given task |

| Course Outcomes<br>At the end of the course student will be able to- | |
|---|---|
| 1 | Apply fundamental principles of Computer Science. |
| 2 | Become specialized in a particular technology domain |
| 3 | Become updated with all the latest changes in technological world. |
| 4 | Communicate efficiently. |
| 5 | Identify, formulate and model problems and find engineering solution based on a systems approach. |
| 6 | Have awareness of the social, cultural, global and environmental responsibility as an engineer. |

## Course Contents

Students have to complete two weeks industrial training program after semester V in Software / hardware Industries, Telecom Sectors, Corporate Offices of their choice with the approval of the Department. At the end of the training student will submit a report as per the prescribed format to the department.

## Course Assessment

This course is mandatory credit based course and the student has to pass the course to be promoted to final year. The student shall make a presentation before a committee constituted by the department which will assess the student based on the report submitted and the presentation made. CIE Marks will be awarded out of 50 and appropriate grades assigned as per the rules and regulations.